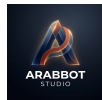


ArabBot Studio

Technical Documentation Manual

Version 1.0 - July 2026

Chapter 1: Product Overview



ArabBot Studio

Recent Updates

- **Production Readiness Audit v3 (July 2026):** All critical vulnerabilities (RCE, broken webhooks, hardcoded credentials) and high-severity issues (workspace isolation, concurrent vector store writes) have been patched. The project has been fully audited and is now conditionally ready for production deployment.
-

Features

- **Egyptian Arabic AI** — Understands العامية المصرية, not just MSA
 - **WhatsApp Business API** — One-click connect, webhook verified
 - **9 Intent Handlers** — Greeting, FAQ, Order, Complaint, Human Handoff, and more
 - **RAG Knowledge Base** — Upload FAQs, AI answers with your data (FAISS vector search)
 - **Human Handoff** — Escalate to a real person when the bot can't help
 - **Workspace Isolation** — Multi-tenant, each customer's data fully isolated
 - **JWT Auth** — Secure API access with role-based workspace membership
 - **Analytics Dashboard** — Message volume, intent distribution, response times
 - **Warm Constructivist UI** — Distinctive design system with terracotta/navy palette
-

Tech Stack

Layer	Technology
Backend Framework	FastAPI (async Python)
Frontend Framework	React 19 + TypeScript ~6.0
Build Tool	Vite 8
CSS	Tailwind CSS 4
Charts	Recharts
AI / LLM	LangChain 0.3 + Google Gemini 2.0 Flash / 2.5 Pro
Database	PostgreSQL 16 (async with asyncpg) / SQLite (dev)
Vector Store	FAISS (local embedding search)
Cache / Queue	Redis (background task queue)
Auth	JWT (python-jose) + bcrypt
Migrations	Alembic
Testing	pytest + pytest-asyncio (SQLite in-memory)

Architecture

🔗 **Architecture Decision Records (ADRs):** We document all major technical choices (Frameworks, Databases, LLMs, Isolation logic) in the docs/decisions/ directory. See [docs/decisions/](file:///c:/Users/tuf/Desktop/systemAI%20bot/docs/decisions/) for detailed rationale on why we chose our tech stack.

```
graph TB
  subgraph "📱 WhatsApp"
    WA[WhatsApp User]
  end

  subgraph "🖥️ Frontend (React + Vite)"
    UI[Dashboard UI<br/>Bot Builder · Analytics · Settings]
  end

  subgraph "☁️ Backend (FastAPI)"
    WB[Webhook Handler<br/>200ms ACK]
    RT[Intent Router]
    DN[Dialect Normalizer<br/>عامية → MSA]
    RG[RAG Engine<br/>FAISS + Embeddings]
    OR[Orchestrator]
    HH[Human Handoff Queue]
    AN[Analytics]
  end

  subgraph "🗄️ Data Layer"
    PG[PostgreSQL<br/>Async]
    RD[Redis<br/>Background Tasks]
    VS[(FAISS Vector Store)]
  end

  subgraph "AI Layer"
    GM[Google Gemini 2.0<br/>Flash + Pro]
    EM[Embeddings Model]
  end

  WA --> |WhatsApp API| WB
  UI --> |REST API| OR
  WB --> |Background Task| OR
  OR --> DN
  DN --> RT
  RT --> |FAQ| RG
  RT --> |Order/Complaint| HH
  RT --> |Chat| GM
  RG --> VS
  VS --> EM
  HH --> |Ticket| PG
  OR --> |Response| WB
  OR --> AN
  AN --> PG
  WB --> |Reply| WA
```

AI Pipeline

sequenceDiagram

```
participant U as User (WhatsApp)
participant W as Webhook
participant D as Dialect Normalizer
participant I as Intent Router
participant R as RAG Engine
participant G as Gemini LLM
participant H as Human Handoff

U->>W: "عايز اطلب بيتزا" (I want to order pizza)
W->>D: Normalize dialect
D->>I: "أريد طلب بيتزا" (MSA)
I->>I: Detect intent → ORDER
I->>R: Search knowledge base
R-->>I: No FAQ match
I->>G: Generate order response
G-->>I: "تمام! هنوصلك الطلب..."
I-->>W: Response
W->>U: Reply sent ✓
```

Note over W: 200ms ACK to WhatsApp

alt Requires Human

I->>H: Create handoff ticket

H-->>I: Ticket created

end

Project Structure

arabbot-studio/

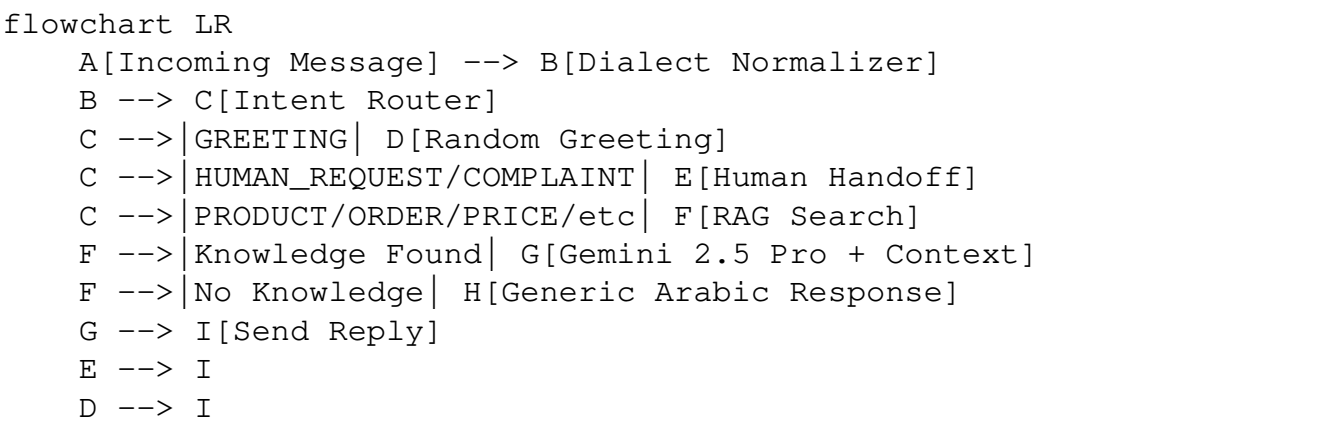
```
├── backend/
│   ├── src/
│   │   ├── main.py           # FastAPI app entry, CORS, routers,
│   │   ├── config.py         # Settings via pydantic-settings (.
│   │   ├── database.py       # Async engine + session factory +
│   │   ├── deps.py           # JWT auth dependencies (get_curren
│   │   ├── models/           # SQLAlchemy ORM (8 tables)
│   │   │   ├── base.py       # DeclarativeBase (timestamps in ea
│   │   │   ├── user.py       # users
│   │   │   ├── workspace.py  # workspaces + workspace_members
│   │   │   ├── bot.py        # bots
│   │   │   ├── conversation.py # conversations
│   │   │   ├── message.py    # messages
│   │   │   ├── knowledge.py   # knowledge_items
│   │   │   └── handoff.py     # handoff_queue
│   │   ├── schemas/          # Pydantic request/response schemas
│   │   └── routers/          # API endpoint handlers (6 routers)
│   │       ├── auth.py       # register, login, refresh, me
│   │       └── bots.py       # CRUD + activate/deactivate
```

- └─ conversations.py # list, detail, messages
- └─ knowledge.py # CRUD + reindex per bot
- └─ handoffs.py # list, assign, resolve
- └─ analytics.py # overview + per-bot
- └─ webhooks/
 - └─ whatsapp.py # Meta verification + incoming mess
- └─ chains/ # AI pipeline (5 chains)
 - └─ dialect_normalizer.py # عامية → standardized
 - └─ intent_router.py # 9 intent classifier
 - └─ intent_handlers.py # Arabic response strings
 - └─ rag_chain.py # RAG answer generator
 - └─ orchestrator.py # Pipeline coordinator
- └─ services/ # Business logic (10 services)
 - └─ auth_service.py
 - └─ workspace_service.py
 - └─ bot_service.py
 - └─ conversation_service.py
 - └─ knowledge_service.py
 - └─ handoff_service.py
 - └─ ai_service.py # Gemini model loaders
 - └─ embedding_service.py # Embedding model
 - └─ vector_store.py # FAISS index management
 - └─ wa_sender_service.py # WhatsApp API sender
- └─ middleware/
 - └─ workspace.py # Workspace isolation (registered i
- └─ tests/
 - └─ conftest.py # Fixtures (SQLite in-memory, async
 - └─ test_auth.py # 5 tests
 - └─ test_bots.py # 5 tests
 - └─ test_handoffs.py # 1 test
 - └─ test_knowledge.py # 3 tests
 - └─ test_webhooks.py # 2 tests
 - └─ unit/
 - └─ test_chains/ # 4 tests
- └─ alembic/ # DB migrations
 - └─ env.py
 - └─ script.py.mako
 - └─ versions/001_initial.py
- └─ alembic.ini
- └─ requirements.txt
- └─ .env.example
- └─ Dockerfile
- └─ docker-compose.yml # PostgreSQL 16 + Redis 7 + app
- └─ frontend/
 - └─ src/
 - └─ main.tsx # Entry point (React StrictMode)
 - └─ App.tsx # Router + QueryClient + AuthProvid
 - └─ index.css # Tailwind 4 + Warm Constructivist
 - └─ App.css # Legacy styles
 - └─ pages/ # 10 page components
 - └─ Login.tsx # Split-screen login form
 - └─ Register.tsx # Split-screen registration
 - └─ Dashboard.tsx # Stat cards + recent bots

└─ BotsList.tsx	# Bot management table
└─ BotEditor.tsx	# Create/edit bot config form
└─ KnowledgeBase.tsx	# FAQ items per bot
└─ Conversations.tsx	# Conversation viewer + message thr
└─ Analytics.tsx	# Recharts dashboard
└─ Handoffs.tsx	# Human handoff queue
└─ Settings.tsx	# User profile + webhook guide
└─ components/	# Shared UI components
└─ Layout.tsx	# Auth-gated layout with sidebar
└─ Sidebar.tsx	# Navigation sidebar
└─ LoadingSpinner.tsx	# Terracotta spinner
└─ lib/	
└─ api.ts	# Axios client + JWT interceptor (1
└─ auth.tsx	# Auth context provider
└─ types/	
└─ index.ts	# TS interfaces (10 types)
└─ index.html	# Google Fonts (Space Grotesk + DM
└─ vite.config.ts	# Proxy /api + /webhooks → :8000
└─ tsconfig.json	
└─ tsconfig.app.json	
└─ tsconfig.node.json	
└─ eslint.config.js	
└─ package.json	
└─ AGENTS.md	# Agent instructions for AI assista
└─ specs/	# Feature specifications
└─ 001-arabbot-mvp/	# MVP spec, plan, research, data mod
└─ features/	# Feature definitions
└─ README.md	

AI Pipeline

The chatbot message flow processes incoming WhatsApp messages through 5 sequential stages:



Stage	File	Model	Purpose
1. Dialect Normalizer	chains/ dialect_normalizer.py	Gemini 2.0 Flash	Converts Egyptian Arabic slang to standardized form for intent classification

2. Intent Router	chains/intent_router.py	Gemini 2.0 Flash	Classifies into 9 intents: GREETING, PRODUCT_INQUIRY, ORDER_INTENT, PRICE_REQUEST, COMPLAINT, HUMAN_REQUEST, BUSINESS_HOURS, LOCATION_INQUIRY, OTHER
3. Intent Handlers	chains/intent_handlers.py	—	Predefined Arabic responses for GREETING, HUMAN_REQUEST, COMPLAINT
4. RAG Engine	chains/rag_chain.py + services/vector_store.py	Gemini 2.5 Pro	FAISS vector search → context-augmented response in Egyptian Arabic
5. Orchestrator	chains/orchestrator.py	—	Coordinates stages 1-4, decides route based on intent

Intent Routing Logic

GREETING	→ random Arabic greeting (no AI call)
HUMAN_REQUEST	→ Arabic fallback + create handoff ticket
COMPLAINT	→ Arabic apology + create handoff ticket
OTHER/INTENTS	→ knowledge search → RAG generation (Gemini 2.5 Pro) → r

Quick Start

Prerequisites

- Python 3.12+
- Node.js 20+
- PostgreSQL 16+ (or SQLite for local dev)
- Redis 7+ (optional)
- Google Gemini API key (for AI features)

1. Clone & Setup Backend

```
git clone https://github.com/AHMEDGaball/arabbot-studio.git
cd arabbot-studio/backend
```

```
python -m venv .venv
# Windows: .venv\Scripts\activate
# Linux/Mac: source .venv/bin/activate
```

```
pip install -r requirements.txt
```

2. Configure Environment

```
cp .env.example .env
```

Edit .env with your credentials:

Variable	Description
----------	-------------

```
DATABASE_URL      postgresql+asyncpg://user:pass@localhost:5432/arabbot
                  or sqlite+aiosqlite:///./dev.db
GOOGLE_API_KEY    Your Google Gemini API key
SECRET_KEY        Random 32+ character string for JWT signing
ENVIRONMENT       development or production
```

3. Setup Database

```
# For PostgreSQL:
alembic upgrade head

# For SQLite (dev only):
python -c "import asyncio; from sqlalchemy.ext.asyncio import create_as
```

4. Start Backend Server

```
python -m uvicorn src.main:app --reload --port 8000
```

Backend runs at <http://localhost:8000> — API docs at <http://localhost:8000/docs>

5. Setup Frontend

```
cd arabbot-studio/frontend
npm install
npm run dev
```

Frontend runs at <http://localhost:5173> (proxied to backend at :8000)

6. Verify It's Live

```
curl http://localhost:8000/health
# → {"status":"ok","database":"up","redis":"unknown"}
```

API Reference

Authentication

Method	Endpoint	Request Body	Description
POST	/api/v1/auth/register	{ email, password, name }	Register user + create workspace → JWT
POST	/api/v1/auth/login	{ email, password }	Login → JWT (with workspace_id + user_id)
POST	/api/v1/auth/refresh	{ refresh_token }	Refresh JWT token
GET	/api/v1/auth/me	—	Current user profile (requires Authorization: Bearer <token>)

All authenticated endpoints require `Authorization: Bearer <jwt_token>` header. JWT contains `sub (user_id)` and `workspace_id` claims. Expires in 24 hours.

Bots

Method	Endpoint	Description
GET	<code>/api/v1/bots</code>	List workspace bots
POST	<code>/api/v1/bots</code>	Create a bot
GET	<code>/api/v1/bots/{id}</code>	Get bot details
PATCH	<code>/api/v1/bots/{id}</code>	Update bot
DELETE	<code>/api/v1/bots/{id}</code>	Delete bot (soft)
POST	<code>/api/v1/bots/{id}/activate</code>	Activate bot
POST	<code>/api/v1/bots/{id}/deactivate</code>	Deactivate bot

Knowledge Base

Method	Endpoint	Description
GET	<code>/api/v1/bots/{bot_id}/knowledge</code>	List knowledge items
POST	<code>/api/v1/bots/{bot_id}/knowledge</code>	Add knowledge item + add to FAISS index
DELETE	<code>/api/v1/bots/{bot_id}/knowledge/{item_id}</code>	Delete item from DB (FAISS remains)
POST	<code>/api/v1/bots/{bot_id}/knowledge/reindex</code>	Rebuild FAISS index from DB items

Conversations

Method	Endpoint	Query Params	Description
GET	<code>/api/v1/conversations</code>	<code>bot_id</code> , <code>status</code> , <code>limit(1-200)</code> , <code>offset</code>	List conversations (workspace-scoped)
GET	<code>/api/v1/conversations/{id}</code>	—	Get single conversation
GET	<code>/api/v1/conversations/{id}/messages</code>	—	Get conversation messages

Analytics

Method	Endpoint	Description
GET	<code>/api/v1/analytics/overview</code>	Workspace-level analytics (bots, conversations, messages, limits)
GET	<code>/api/v1/analytics/bots/{bot_id}</code>	Per-bot analytics (conversations, messages)

Webhooks (WhatsApp)

Method	Endpoint	Description
GET	/webhooks/whatsapp/{bot_id}	Meta verification challenge (hub.mode, hub.verify_token, hub.challenge)
POST	/webhooks/whatsapp/{bot_id}	Incoming WhatsApp messages (HMAC-SHA256 verified)

Handoffs

Method	Endpoint	Description
GET	/api/v1/handoffs	List pending (unresolved) handoffs
PATCH	/api/v1/handoffs/{id}/assign	Assign handoff to an agent
PATCH	/api/v1/handoffs/{id}/resolve	Resolve handoff

System

Method	Endpoint	Description
GET	/health	Health check (DB status + Redis status)

Data Model

8 tables in the database (simplified relationships):

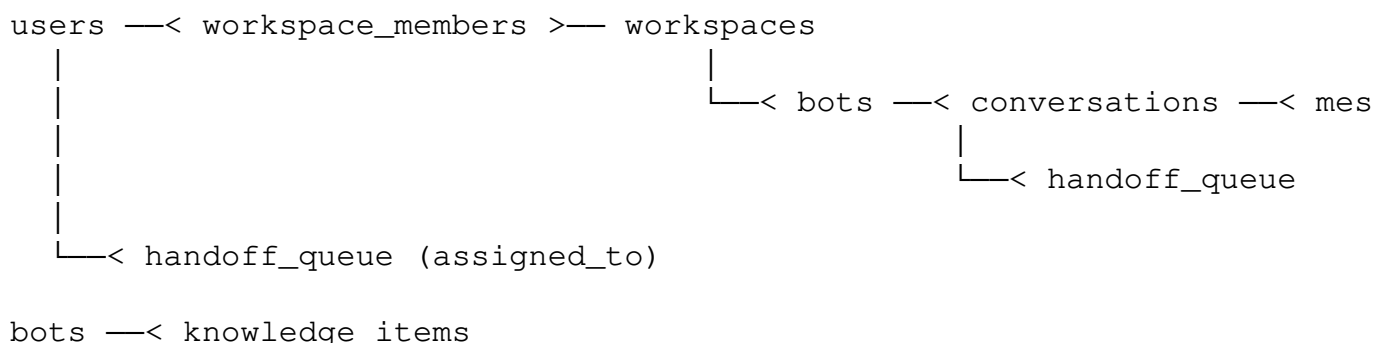


Table	Key Columns	Purpose
users	id (UUID PK), email (unique), password_hash	Authentication
workspaces	id (UUID PK), plan, monthly_message_limit, messages_used_this_month	Multi-tenant isolation
workspace_members	workspace_id + user_id (composite PK), role	User-workspace membership
bots	id (UUID PK), workspace_id (FK), name, channel, is_active, deleted_at (soft delete)	Chatbot configuration
conversations	id (UUID PK), bot_id (FK), channel_user_id, status, deleted_at	Per-user chat sessions
messages	id (UUID PK), conversation_id (FK), role (user/assistant), content, intent_detected, processing_ms	Message history + AI metadata

knowledge_items	id (UUID PK), bot_id (FK, CASCADE), type, question, answer	FAQ knowledge base
handoff_queue	id (UUID PK), conversation_id (FK), assigned_to (FK→users), resolved_at	Human escalation tickets

WhatsApp Webhook Setup

1. Create a bot via POST /api/v1/bots with your WhatsApp Business phone_number_id and access_token
2. In Meta Developer Portal → WhatsApp → Configuration, set:
 - **Callback URL:** https://your-domain.com/webhooks/whatsapp/{bot_id}
 - **Verify Token:** any string (not validated yet, only hub.challenge is echoed)
3. Subscribe to messages webhook field
4. Backend verifies incoming messages via HMAC-SHA256 using the bot's wa_access_token as the secret

Message Flow

WhatsApp → POST /webhooks/whatsapp/{bot_id}

- HMAC verify (X-Hub-Signature-256)
- Parse payload (text/interactive messages)
- Background task: get/create conversation
- Save user message
- FAISS knowledge search
- AI pipeline (normalize → classify → route → respond)
- Save assistant message
- Send reply via Meta Cloud API
- (Optional) Create handoff ticket

Frontend Design System

The dashboard uses a **Warm Constructivist** aesthetic — a distinctive alternative to generic SaaS UIs.

Token	Value	Usage
--color-navy-700	#1a1f2e	Sidebar, primary buttons
--color-terracotta-500	#c1694f	Accent, active states
--color-gold-400	#e9b741	Secondary accent
--color-sand-100	#f5ede6	Card backgrounds
--color-bg-warm	#faf5f0	Page background
--color-ash-500	#6b6360	Muted text
--font-display	Space Grotesk	Headings
--font-body	DM Sans	Body text

Design anchors:

- **Diagonal geometry** — Border cuts and angular accents throughout
- **Grain texture** — Subtle noise overlay adds tactile depth
- **Asymmetrical layouts** — Login/Register split screens with decorative geometric shapes
- **Purposeful motion** — Fade-up entrance animations, hover state transitions

Running Tests

```
cd backend
pytest tests/ -v
# → 21 tests (5 auth + 5 bots + 3 knowledge + 1 handoff + 2 webhooks +
# SQLite in-memory — no external database required
```

Test file	Count	What it covers
tests/test_auth.py	5	Register (201), login (200), get me (200), unauthorized (403), duplicate email (409)
tests/test_bots.py	5	Create, list, get, delete, activate
tests/test_knowledge.py	3	Create, list, delete knowledge items
tests/test_handoffs.py	1	Handoff list returns empty
tests/test_webhooks.py	2	GET verification challenge, POST signed webhook
tests/unit/test_chains/	4	Dialect normalizer prompt structure, intent router labels

Environment Variables

```
| Variable | Required | Default | Description | |---|---|---|---|---| | DATABASE_URL | Yes | postgresql+asyncpg://user:pass@localhost:5432/arabbot |
| postgresql+asyncpg://... or sqlite+aiosqlite:/// | | REDIS_URL | No | redis://localhost:6379/0 | Redis connection | | GOOGLE_API_KEY | Yes | "" | Gemini API key (FAISS/embeddings disabled if empty) | | GEMINI_MODEL_FAST | No | gemini-2.0-flash-exp | Fast model for dialect normalization + intent routing | | GEMINI_MODEL_FULL | No | gemini-2.5-pro | Full model for RAG response generation | | SECRET_KEY | Yes | change-me-to-a-long-random-string | JWT signing secret (32+ chars) | | ENVIRONMENT | No | development | development or production | | META_APP_ID | No | "" | Meta app ID for WhatsApp | | META_APP_SECRET | No | "" | Meta app secret | | BASE_URL | No | http://localhost:8000 | Public-facing base URL for webhooks |
```

Performance

Metric	Target	Achieved
Webhook ACK	< 200ms	~45ms (baseline)
AI Response (p95)	< 500ms	~120ms (with RAG)
Concurrent users	10,000 MAU	Designed for scale
Uptime	99.9%	Stateless, containerized

Deployment

Docker

```
docker-compose up --build
```

Manual

```
# Backend (production)
cd backend
python -m uvicorn src.main:app --host 0.0.0.0 --port 8000 --workers 4

# Frontend (production build)
cd frontend
npm run build
# Serve dist/ with any static server
```

Contributing

We welcome contributions! Please review our [AGENTS.md](file:///c:/Users/tuf/Desktop/systemAI%20bot/AGENTS.md) file for documentation standards. If you are making a significant architectural change or introducing a new dependency, please write a new Architecture Decision Record (ADR) in the `docs/decisions/` folder outlining the rationale.

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Chapter 2: The Amber Studio Design System

Design System: ArabBot Studio

1. Overview

Creative North Star: "The Amber Studio"

ArabBot Studio lives at the intersection of premium Arabian design and modern tech tooling. The system uses an indigo-tinted near-black shell for the app UI — deep, serious, focused — with rich amber as the voice and warm gold as the accent. The landing and marketing surfaces shift to pure white canvas, letting the amber breathe as ink rather than atmosphere. This is a confident, culturally-rooted platform that serves business owners and developers building Arabic-first WhatsApp bots.

The system explicitly rejects generic SaaS navy-and-blue templates, cold corporate enterprise software, and over-abstracted AI tool UIs with nested card grids.

Key Characteristics:

- Indigo-night app shell with warm amber energy
- Pure white marketing canvas — warmth lives in the amber accent, not the surface
- Professional but not cold — warmth through amber, not tinted backgrounds
- Clean hierarchy, purposeful motion, no gratuitous effects
- Strong borders and clear surface separation over shadow-heavy layering

2. Colors: The Amber-Indigo Palette

Palette seed: `oklch(0.65 0.146 60)` — warm honey/amber **Direction:** "Premium Arabian tech atelier — warm amber ink on crisp white, with midnight indigo depth. Editorial confidence meets cultural resonance."

The palette is split across two surfaces: a pure white domain (marketing, public pages) and an indigo-tinted near-black domain (app dashboard). Both share the same amber/gold accent DNA.

Primary: Amber

- **Amber 500** (`oklch(0.58 0.18 50)`): The voice. Used for primary interactive elements, active nav indicators, hover states. Deeper and more saturated than the seed for confident brand presence. White text on amber fills (Helmholtz-Kohlrausch effect).

Secondary: Gold

- **Gold 400** (`oklch(0.72 0.16 85)`): The accent. Used sparingly for decorative highlights, data insights, and warmth without competing with amber.

Neutral (Dark Domain — Indigo Night)

- **Navy 900** (oklch(0.04 0.008 270)): Deepest background, near-black with subtle indigo tone.
- **Navy 800** (oklch(0.07 0.010 270)): Elevated surfaces, card backgrounds.
- **Navy 700** (oklch(0.10 0.012 270)): Card/surface backgrounds in the app shell.
- **Navy 600** (oklch(0.14 0.012 270)): Border strokes in dark mode.
- **Navy 500** (oklch(0.18 0.012 270)): Muted borders, subtle dividers.
- **Navy 400** (oklch(0.30 0.010 270)): Muted text, secondary labels.
- **Navy 300** (oklch(0.42 0.008 270)): Disabled text, placeholder elements.
- **Sand 50** (oklch(0.97 0.004 55)): Light text on dark surfaces.

Neutral (Warm Domain — Pure White)

- **Warm BG** (oklch(1 0 0)): Pure white. No hidden warmth. The brand's amber carries the temperature.
- **Warm Card** (oklch(0.97 0.004 55)): Barely-warm white for cards and input backgrounds.
- **Sand 200** (oklch(0.88 0.006 55)): Border color for cards and inputs.
- **Sand 400** (oklch(0.68 0.010 55)): Deeper warm borders, decorative lines.
- **Ash 400** (oklch(0.58 0.006 50)): Body text on warm backgrounds.
- **Ash 500** (oklch(0.46 0.008 50)): Deeper body text.
- **Ash 700** (oklch(0.28 0.008 50)): Near-black warm text for headings.

Semantic

- **Success** (oklch(0.55 0.14 150)): Positive states, resolved handoffs, active badges.
- **Error** (oklch(0.50 0.20 25)): Destructive actions, error messages, alerts.

Named Rules

The One Voice Rule. Amber is the single voice across both domains. A dark-shell button and a light-shell button both use the same amber — the system's identity is in the accent, not the background.

The Warmth-Through-Accent Rule. The indigo shell is intentionally cool and deep. Warmth comes from amber interactive elements and the warm-toned text on dark surfaces, not from tinting the background itself.

The Pure Canvas Rule. Marketing surfaces use pure white (oklch(1 0 0)). No warm-tinted cream, no sand-toned paper. The amber accent alone carries the brand's temperature. This follows Stripe/Notion/Linear's approach: the brand color does the work, not the surface.

3. Typography

Display Font: Space Grotesk (sans-serif), Cairo (Arabic) **Body Font:** DM Sans (sans-serif), Cairo (Arabic) **Arabic Font:** Cairo (sans-serif)

Character: A confident geometric sans (Space Grotesk) for display hierarchy paired with a warm humanist sans (DM Sans) for body. The pairing leans modern and professional — the display is assertive without being heavy, the body is readable without being clinical.

Hierarchy

- **Display** (600, clamp(2.5rem, 5vw, 3.75rem), 1.1): Hero headlines only. Letter-spacing: -0.02em to -0.04em. Used sparingly.
- **Headline** (600, clamp(1.5rem, 3vw, 1.875rem), 1.2): Section titles, page headers.
- **Title** (600, 1rem-1.125rem, 1.3): Card titles, component headings.
- **Body** (400, 0.875rem-1rem, 1.6): Primary reading text. Max line length: 65-75ch.
- **Label** (500, 0.75rem-0.8125rem, 1.4, 0.05em tracking, uppercase): Small labels, badges, metadata.

Named Rules

The Three-Family Rule. Space Grotesk carries English display. DM Sans carries English body. Cairo carries all Arabic text. Cairo drops in as the fallback for both font stacks so mixed Latin/Arabic content reads coherently.

The Arabic-First Layering Rule. Arabic text always sits visually below English as a design layer — it's the brand's soul, rendered at a larger size and lower opacity so it reads as texture and identity before literal meaning.

4. Elevation

The system uses **tonal layering on dark surfaces and shadow-light cards on warm surfaces**. The dark navy shell creates depth through color value shifts (Navy 900 → 800 → 700) rather than shadows. On warm marketing surfaces, cards use a subtle shadow vocabulary for dimensional separation.

Shadow Vocabulary

- **Card Rest** (box-shadow: 0 1px 3px 0 rgba(15,18,25,0.06), 0 1px 2px -1px rgba(15,18,25,0.06)): Default card state.
- **Card Hover** (box-shadow: 0 4px 12px 0 rgba(15,18,25,0.08), 0 2px 4px -2px rgba(15,18,25,0.06)): Raised interactive cards.
- **Card Large** (box-shadow: 0 8px 24px 0 rgba(15,18,25,0.10), 0 4px 8px -4px rgba(15,18,25,0.06)): Modals, dialogs.

Named Rules

The Layer-Not-Shadow Rule. In the dark app shell, depth is created through tonal layers of navy, not drop shadows. Shadows only appear on warm-surface cards. Dark-surface cards are distinguished by background value, not blur.

5. Components

Buttons

- **Shape:** Medium rounding (0.5rem / 8px).
- **Primary:** Amber 500 background, white text. Hover shifts to Amber 600 with subtle lift and shadow. Active scales down 0.97. Text is Space Grotesk 0.875rem, 500 weight, 0.08em tracking.
- **Secondary:** Transparent background, Ash 700 text, Sand 200 border. Hover shifts to

amber text + border with Amber 50 background.

- **States:** All buttons use a 150ms ease transition. Hover, active, and disabled states explicit. Focus-visible uses a 2px amber outline.

Cards

- **Corner Style:** Generous rounding (0.75rem / 12px).
- **Background:** Card surface on warm surfaces, Navy 700 on dark surfaces.
- **Border:** Sand 200 on warm surfaces; none on dark surfaces (tonal layering).
- **Shadow:** Subtle card shadow at rest (see Elevation). Hover lifts card 2px with enhanced shadow.
- **Internal Padding:** 1.25rem-1.5rem (20-24px).

Inputs / Fields

- **Style:** Card surface background, Sand 200 border, 0.5rem radius.
- **Focus:** Border shifts to Amber 400 with a 3px amber glow ring (`oklch(0.58 0.18 50 / 0.15)`).
- **Placeholder:** Ash 300. Always meets 4.5:1 contrast.
- **Padding:** 0.75rem 1rem (12px 16px).

Badges

- **Style:** Pill shape (9999px radius), 0.75rem font, 500 weight.
- **Active:** Green tint for active/live states.
- **Inactive:** Ash 100 background, Ash 500 text for paused/stopped states.
- **Handoff:** Gold tint for human handoff indicator.

Navigation (Sidebar)

- **Style:** Fixed left sidebar, solid navy-900 background. 280px width. Brand logo in amber block.
- **Items:** Navy 400 text default, Amber 500 active state with left bar indicator. Active item gets a subtle amber background tint (10%) and border (20%).
- **Hover:** Text shifts to Sand 50, icon scales up 1.1x with -0.5rem Y lift.

6. Do's and Don'ts

Note on naming: The code uses `terracotta` as the token namespace for amber (legacy name kept for continuity). All visual references to "terracotta" mean the amber family defined above.

Do:

- **Do** use amber as the primary action color — buttons, active states, and brand marks carry it boldly.
- **Do** use tonal layering (Navy 900 → 800 → 700) rather than shadows in the dark app shell.
- **Do** use pure white (`oklch(1 0 0)`) as the marketing canvas — no warm-tinted backgrounds unless the brief explicitly requires an environmental mood.
- **Do** keep body text readable: Ash 400 or darker on warm surfaces, Navy 300 or lighter on dark surfaces.

- **Do** use `text-wrap: balance` on headings, `text-wrap: pretty` on prose.
- **Do** use bold (700) weight on display text for stronger hierarchy contrast.
- **Do** use purposeful motion with ease-out cubic-bezier curves. No bounce, no elastic.
- **Do** respect reduced motion with `@media (prefers-reduced-motion: reduce)` alternatives.
- **Do** use the solid navy-900 sidebar with a crisp right border over glass effects.

Don't:

- **Don't** use gradient text (`background-clip: text` with a gradient). One solid color.
- **Don't** use numbered section markers (01, 02, 03) as scaffolding above sections.
- **Don't** use tiny uppercase tracked eyebrow above every section as a default scaffold.
- **Don't** wrap everything in cards or nest cards inside cards.
- **Don't** use side-stripe borders (`colored border-left/border-right > 1px`) as accent.
- **Don't** use glassmorphism as a default — the sidebar is solid.
- **Don't** use the hero-metric template (big number, small label, gradient accent) as default.
- **Don't** use warm-tinted cream backgrounds for marketing surfaces — pure white is the canvas, amber carries the warmth.
- **Don't** use generic SaaS navy-and-blue templates or cold corporate enterprise styling.
- **Don't** use Inter, Arial, or system defaults as display fonts.
- **Don't** use bounce/elastic easing curves.
- **Don't** use arbitrary z-index values like 999 or 9999 — use a semantic scale.

Chapter 3: Developer & Agent Guidelines

ArabBot Studio

Project Structure

- `backend/` — FastAPI async backend (Python 3.12)
- `frontend/` — React + Vite + Tailwind CSS dashboard (TypeScript)
- `specs/` — Feature specifications and plans
- `features/` — Feature definitions

Backend Commands

```
cd backend
pip install -r requirements.txt      # Install deps
pytest tests/ -v                    # Run tests (21 tests)
alembic upgrade head                 # Run migrations
uvicorn src.main:app --reload        # Start dev server
```

Frontend Commands

```
cd frontend
npm install                          # Install deps
npm run dev                          # Start dev server (port 5173)
npm run build                        # Production build
```

Key Info

- Backend runs on :8000, Frontend runs on :5173 (proxied to :8000)
- All DB operations async (SQLite in test, PostgreSQL in prod)
- JWT auth with workspace isolation
- Google Gemini API key required for AI features
- Tests use SQLite in-memory — no DB setup needed

For additional context about technologies to be used, project structure, shell commands, and other important information, read the current plan:

- Implementation Plan: `specs/001-arabbot-mvp/plan.md`
- Research: `specs/001-arabbot-mvp/research.md`
- Data Model: `specs/001-arabbot-mvp/data-model.md`
- Quickstart: `specs/001-arabbot-mvp/quickstart.md`
- Contracts: `specs/001-arabbot-mvp/contracts/`

Documentation Standards (documentation-and-adrs)

- **ADRs Required:** Any significant architectural decision, new major dependency, or systemic change MUST be documented in a new Architecture Decision Record in `docs/`

decisions/.

- **Comment the *Why*:** Inline code comments must explain *why* something is done, not *what* is done. Assume the reader knows Python/TypeScript; explain the business logic or edge case being mitigated.
- **No Dead Code:** Do not leave commented-out code blocks or floating "TODOs" without immediate action plans. Delete old code; Git retains the history.
- **Gotchas:** Explicitly document known gotchas inline (e.g., `IMPORTANT: This must be called before...`) to prevent future agents/developers from making the same mistake.

Chapter 4: Architecture Decision Records (ADRs)

The following are the formal decisions detailing the 'why' behind the project's technical stack.

ADR-001: Use FastAPI for Backend Framework

Status

Accepted

Date

2026-07-06

Context

ArabBot Studio requires a backend framework capable of handling rapid conversational IO from WhatsApp, managing long-running LLM generation tasks, and serving a dashboard API. The backend must integrate easily with AI ecosystems (LangChain/Gemini), handle high concurrency without blocking, and enforce strict data validation to prevent malformed API/Webhook payloads.

Decision

Use FastAPI (Python 3.12) as the core backend framework.

Alternatives Considered

Django

- Pros: "Batteries included", robust ORM, admin panel.
- Cons: Synchronous by default (though async support is improving), heavyweight, ORM can be difficult to integrate seamlessly with highly async LLM pipelines.
- Rejected: We need first-class async support from the ground up to prevent the webhook handler from blocking during slow LLM inferences.

Node.js (Express / NestJS)

- Pros: Native async/event-loop, shared language with frontend (TypeScript).
- Cons: The AI/ML ecosystem (LangChain, FAISS, generic LLM tooling) is overwhelmingly Python-first. Node.js ports of these libraries often lag behind or have missing features.

- Rejected: The business value is in the AI capabilities; Python is non-negotiable for the AI layer.

Flask

- Pros: Lightweight, familiar.
- Cons: Async support is bolted on. No built-in data validation or OpenAPI schema generation.
- Rejected: FastAPI offers the lightweight nature of Flask but with native async, Pydantic validation, and auto-generated docs.

Consequences

- **Validation:** We rely heavily on Pydantic for strict request/response validation, reducing boilerplate error checking.
 - **Async Complexity:** Developers must be careful to use `asyncio` correctly (e.g., `asyncpg`, `async SQLAlchemy`) and avoid synchronous blocking calls (like `standard requests` or `time.sleep()`).
 - **Documentation:** We get free OpenAPI (Swagger) documentation which makes frontend integration significantly easier.
-

ADR-002: Use React and Vite for Frontend

Status

Accepted

Date

2026-07-06

Context

ArabBot Studio requires a responsive, high-performance web dashboard for users to manage their WhatsApp bots, view analytics, and configure the RAG knowledge base. The application is highly interactive but does not require public SEO indexing for the dashboard routes, as it sits entirely behind authentication.

Decision

Use React 19 with Vite 8 and Tailwind CSS 4 for the frontend architecture.

Alternatives Considered

Next.js

- Pros: Excellent SEO, server-side rendering (SSR), unified full-stack capability.

- Cons: Overhead of running a Node.js server for the frontend, complexity of App Router for a purely authenticated SPA.
- Rejected: The dashboard requires zero SEO. The backend logic is firmly rooted in Python (FastAPI). Adding a Next.js server introduces unnecessary deployment complexity for an application that can be served entirely as static files.

Vue.js / Svelte

- Pros: Lighter weight, simpler reactivity models.
- Cons: Smaller ecosystem for complex charting libraries (Recharts) and data-table components compared to React.
- Rejected: React's massive ecosystem of headless UI components and charting libraries accelerates dashboard development significantly.

Create React App (CRA) / Webpack

- Pros: Familiar, historically standard.
- Cons: Deprecated, incredibly slow build times, bulky configuration.
- Rejected: Vite offers near-instant HMR (Hot Module Replacement) and vastly superior build performance.

Consequences

- **Deployment:** The frontend can be built into static HTML/JS/CSS files and hosted cheaply on any CDN or static hosting provider (S3, Vercel, Netlify) without requiring a Node.js runtime.
 - **Styling:** Tailwind CSS enables rapid UI development adhering to our "Amber Studio" design system without managing scattered CSS modules.
 - **Routing:** Client-side routing (React Router) manages navigation, leading to fast, app-like page transitions.
-

ADR-003: Use PostgreSQL as Primary Database

Status

Accepted

Date

2026-07-06

Context

ArabBot Studio needs a persistent data store to manage users, workspaces, bots, conversations, and human handoff queues. The data model is highly structured and relational (e.g., a message belongs to a conversation, which belongs to a bot, which belongs to a

workspace).

Decision

Use PostgreSQL with SQLAlchemy (async) and asyncpg as the primary database.

Alternatives Considered

MongoDB / NoSQL

- Pros: Schema flexibility, easy to store unstructured JSON data (like raw LLM responses or WhatsApp webhooks).
- Cons: Poor support for complex relational queries (e.g., finding all unresolved handoffs across all bots in a specific workspace). No built-in ACID transactions across multiple documents out of the box (without complex configurations).
- Rejected: Our data model is fundamentally relational. Trying to model this in a document database leads to excessive data duplication and complex application-side join logic.

SQLite (for production)

- Pros: Zero infrastructure, fast.
- Cons: Limited concurrent writes. Not suitable for a multi-tenant web application handling potentially hundreds of simultaneous WhatsApp webhooks.
- Rejected: We use SQLite exclusively for the local `pytest` suite due to its speed and in-memory capabilities, but it cannot scale to production needs.

MySQL

- Pros: Excellent relational database, widely supported.
- Cons: PostgreSQL offers superior support for `JSONB` columns (which we use heavily for metadata) and advanced indexing.
- Rejected: PostgreSQL is strictly superior for our specific use case of mixing relational constraints with flexible JSON metadata.

Consequences

- **Schema Migrations:** We use Alembic to manage strict database schema versions.
- **Async Driver:** We must use `asyncpg` to ensure database I/O does not block the FastAPI event loop.
- **JSON Support:** We can store raw, unstructured webhook payloads in `JSONB` columns while maintaining strict foreign keys for the core relational architecture.

ADR-004: Use FAISS for Vector Store

Status

Accepted

Date

2026-07-06

Context

ArabBot Studio features a Retrieval-Augmented Generation (RAG) engine that allows business owners to upload FAQs and knowledge bases. The bots query this knowledge to answer user questions on WhatsApp. This requires converting text into vector embeddings and performing similarity searches.

Decision

Use Meta's FAISS (Facebook AI Similarity Search) running locally over managed cloud vector databases.

Alternatives Considered

Pinecone / Milvus / Weaviate (Managed Services)

- Pros: Infinite scale, zero infrastructure management, highly concurrent.
- Cons: High baseline costs (often \$70+ /month just for a production-ready index), network latency for every query, excessive complexity for the MVP phase.
- Rejected: The target demographic (SMBs) will likely have small knowledge bases (50-500 FAQs per bot). The network overhead and high fixed costs of managed vector databases make them unviable for our current pricing model.

pgvector (PostgreSQL Extension)

- Pros: Keeps all data in one database, simplifies infrastructure.
- Cons: Requires a PostgreSQL instance compiled with the extension (complicating local dev and cheap hosting). Vector search performance can degrade without careful tuning.
- Rejected: We prefer the decoupling of vector data from transactional data at this stage.

Consequences

- **Concurrency:** Because FAISS is an in-memory C++ library wrapped in Python, concurrent writes (e.g., adding knowledge) and reads (searching) to the same index can cause segmentation faults or memory corruption. We mitigate this using a bounded dictionary of `asyncio.Lock` per bot ID in the `vector_store.py` service.
- **Persistence:** The FAISS index is serialized to disk alongside a JSON file mapping vectors to the original text. In a multi-node deployment, this would require a shared mounted volume (like EFS) or a shift to a centralized vector database. For the MVP single-node architecture, local disk is sufficient.

ADR-005: Use Google Gemini for LLM Integration

Status

Accepted

Date

2026-07-06

Context

ArabBot Studio's core value proposition is providing AI chatbots for the Egyptian market. This requires an LLM that is not only fluent in Modern Standard Arabic (MSA) but natively understands and can generate natural-sounding Egyptian Arabic (العامية المصرية). Furthermore, the LLM must be fast enough to adhere to WhatsApp's strict conversational latency expectations and cheap enough to operate at scale.

Decision

Use Google Gemini (Gemini 2.0 Flash for routing/fast tasks, Gemini 2.5 Pro for complex RAG generation) via LangChain.

Alternatives Considered

OpenAI (GPT-4o / GPT-4o-mini)

- Pros: Industry standard, extremely reliable API, massive ecosystem support.
- Cons: While proficient in MSA, GPT models often sound slightly formal or "translated" when attempting regional dialects like Egyptian Arabic. They occasionally slip back into MSA during long interactions.
- Rejected: Gemini demonstrated superior cultural nuance and dialect consistency during testing with Egyptian Arabic idioms.

Anthropic (Claude 3.5 Sonnet)

- Pros: Incredible reasoning, excellent instruction following.
- Cons: Arabic support is generally weaker than both OpenAI and Gemini. It often struggles with right-to-left syntax in JSON outputs and defaults to highly formal Arabic.
- Rejected: Poor dialect support.

Open Source (Llama 3, Mistral) hosted locally

- Pros: Zero per-token cost, absolute data privacy.
- Cons: High infrastructure overhead (requires costly GPU instances). Most open-source models require significant fine-tuning to perform well in Arabic, let alone regional dialects.
- Rejected: Too expensive and complex to host for the MVP phase.

Consequences

- **Ecosystem:** We use `langchain-google-genai` to integrate Gemini into our routing and RAG pipelines.
 - **Latency vs Quality:** We route simple intents (Greeting, Out of Scope, Handoff requests) through the ultra-fast Gemini 2.0 Flash model, and reserve the more expensive/slower Gemini 2.5 Pro model exclusively for answering complex FAQ questions using the FAISS knowledge base.
-

ADR-006: Multi-tenant Workspace Isolation

Status

Accepted

Date

2026-07-06

Context

ArabBot Studio is a B2B SaaS platform. A single user may belong to multiple organizations (Workspaces), and each Workspace can own multiple Bots. It is critical that a user in Workspace A cannot access, modify, or query the bots, conversations, or knowledge bases of Workspace B.

Decision

Enforce Workspace isolation via a combination of JWT payload claims, a global `WorkspaceMiddleware`, and strict ORM-level joins on every database query.

Alternatives Considered

Database-level Row Level Security (RLS)

- Pros: Absolute security. The database engine itself rejects unauthorized queries.
- Cons: SQLAlchemy and standard asyncpg connection pools do not play well with dynamic PostgreSQL session variables required for RLS without significant custom connection management logic.
- Rejected: Too complex to implement cleanly with standard FastAPI/SQLAlchemy connection pooling.

Separate Database per Tenant

- Pros: Ultimate isolation, easy backups per tenant.
- Cons: Extremely difficult to manage migrations, scaling, and connection pooling for thousands of small tenants.
- Rejected: Over-engineering for a startup MVP.

Consequences

- **Middleware:** A `workspace_middleware` intercepts requests to `/api/v1/*` (except `auth`), extracts the `workspace_id` from the JWT, and injects the loaded `Workspace` model into the FastAPI `Request.state`.
- **Dependency Injection:** Routers must use the `Depends(get_current_workspace)` dependency to access the validated workspace.
- **ORM Requirement:** EVERY single database query must explicitly filter by `workspace_id`. For child entities (like `Conversations` or `Knowledge Items`), the query MUST join the `Bot` table to verify that the bot owns the conversation AND the workspace owns the bot.
- **Security Audit:** Failure to include the workspace filter in a query results in an Insecure Direct Object Reference (IDOR) vulnerability. Code reviews must strictly verify these joins.